

2026-27

CONCEPTS | CLINICALS | MCQS

FORENSIC MEDICINE

- Truth Lies in Details.
- IPC sections, poisons & medico-legal essentials.
- Image-based & case-based questions covered.



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IPC- Indian penal code → definition and punishment of crime

Crpc- Criminal Procedure Court → Any procedure for punishment of criminal

IEA- Indian Evidence act

Oral evidence is more valid because of cross examination.

FMGE 25

Types of witness

FMGE 23

1. Common witness

2. Hearsay witness- not valid except dying declaration (32 IEA) and expert witness (45 IEA).

Types of court in India

FMGE 24

Court	Punishment	Fine
Supreme court	Death sentence	Unlimited fine
High court	Death sentence	Unlimited fine
Session court	Death sentence	Unlimited fine
Additional session court	Death sentence	Unlimited fine
Assistant session court	10 yr	Unlimited fine
Chief judicial magistrate court	7 yr	Unlimited fine
1st class judicial magistrate court	3 yr	10k
2nd class judicial magistrate court	1 yr	5k

Juvenile → <18 yr

Juvenile court → 1st class judicial magistrate court Jail for 3 yrs → rehabilitation

Only the president, supreme court and high court has power to forgive the death sentence. The President has the power of Amnesty.

SECTION 184 - ANY POLICE OFFICER CAN REQUEST A MEDICAL EXAMINATION OF THE ACCUSED TO BE DONE BY RMP IN CERTAIN CASES. (INCLUDING RAPE CASES)

Magistrate → 176 CrPc Judicial magistrate**FMGE 25,24,23**

1. Police custody death
2. Police firing death
3. Police interrogation death

Executive magistrate

1. dowry death
2. Mental asylum
3. exhumation

Leading que- yes/no

- ask in cross examination (145 IEA)
- toughest and has no time limit

FMGE 25

Dying declaration	Dying deposition
32 IEA	-
Can be in written and oral form	-
Taken by magistrate (best)	Magistrate only
Only valid if witness dies	-
Oath, leading que not ask	Oath, cross examination asks

IPC 84- McNaughten rule**FMGE 24,23**

- defect of reason
- unsoundness of mind
- not responsible for crime

SECTION 367- UNSOUNDNESS OF MIND**Sexual perversion****NEETPG 24, FMGE26**

1. Sadism- give pain to partner
2. Masochism- suffering pain

Excessive sexual desire-

In male- satyriasis

In female- nymphomania

Fetichism-It is compulsive act for the use of certain object or a body part as a stimulus for the attainment of sexual gratification irrespective of desire

Transvestism- It is intense sexual desire from cross dressing which may manifest as urges, desires or behaviors.

Frotteurism- It is an act of touching or rubbing one's genitals against another person without the consent in order to attain the sexual pleasure or orgasm

Scatologia- phone call sex

Voyeurism/ scotophilia- watching nude, sex

EONISM/TRANSVESTISM : obtained by wearing a dress of opposite sex .

Electra complex- girl attracted to a parent of the opposite sex.

Oedipus complex- son attracted to a parent of the opposite sex.

Pharon complex- brother attracted to the sister

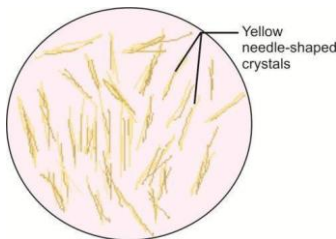
Trace evidence in rape

SEMEN

NEETPG 24,23,FMGE 24

1. Sperm

a)Spermine - Barberios test



b)Choline- florence test

2. Seminal fluid- enzymes

a. Acid phosphatase - best test if sperm is absent

b. LDH- most specific

Blood test

Screening Test

1. Benzidine - blue color

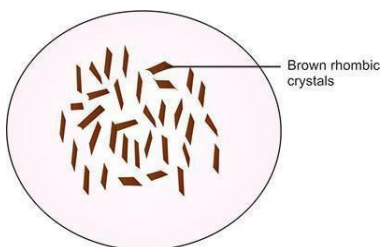
2. Phenolphthalein - pink color

3. Ortho toluidine - bluish green

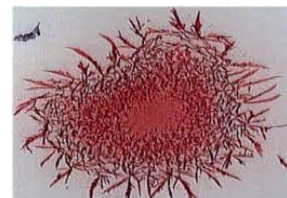
Specific test

1. Teichman - Brown, rhombic dark crystals

Chemical used → NaCl + GAA



2. Takayama- pink feathery crystals



Chemical used → glucose + pyridine + NaOH
3. Spectroscopy- most specific

POCSO ACT

NEETPG 25

Protection of child against sexual offense (applicable if a child <18 years is involved)

POSTHUMOUS CHILD

NEETPG 24

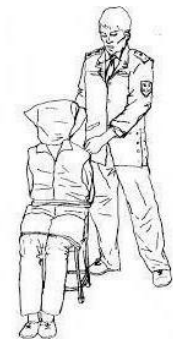
Child born after the death of parents

TWIN PREGNANCY

SUPERFECUNDATION	SUPERFETATION
Occur in 1 menstrual cycle	Occur in 2 menstrual cycle
2 ova fertilised in 2 different act of coitus	2 ova fertilised in 2 different actt of coitus

Types of torture

1. Dry submarine



2. Wet submarine

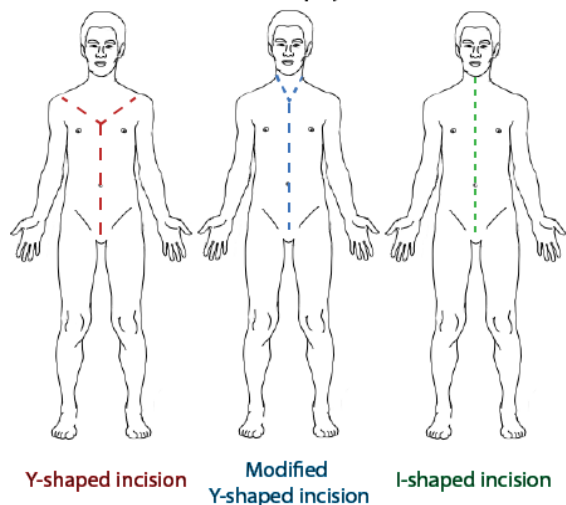


3. Jackknife position- dies due to positional asphyxia



Postmortem incision types

Common Autopsy Incisions



Techniques

NEETPG 23

1. Rokitansky- in-situ - Done in infants
2. Virchow - most common
 - One- one organ removal
3. Ghon's - En- bloc removal
4. Lettule - En- mass removal
5. Virtual- blood less and minimal invasive technique. (CT , MRI)

FMGE 2026

Body cavity to be opened first

Thoracic cavity opened first

Special cases

a. Newborn → Abdomen is to be opened first

b. Poisoning → Cranium is opened first

c. Asphyxial deaths

cranium → thoracic → abdomen → neck

EXHUMATION

Digging out of the dead body in presence of magistrate

Done in early morning

Arsenic present in the soil can go into dead body → postmortem imbibition

BNS

NEETPG 23

20 BNS	A child below 7 years of age is not punishable or liable for any crime
21 BNS	Child b/w 7-12 years can be punishable or liable for any crime but

	depends on mental maturity
22 BNS	Insane are not responsible or liable for the crime
23 BNS	Involuntary intoxication
24 BNS	Voluntary intoxication
26 BNS	Dying declaration
30 BNS	Life threatening or emergency situation
88 BNS	With the consent of the mother
89 BNS	Without the consent of the mother
90 BNS	If mother dies because of complications
91 BNS	Killing child or fetus above 28 weeks of gestation
92 BNS	Intention to kill the mother but the fetus gets killed
106 BNS	Medical negligence
80 BNS	Dowry death

Medical negligence- malpractice

Civil- tort	criminal
No skill	Gross negligence
No care	Willful negligence
Mild damage to patient	death , severe damage
Doctor only has to pay money only	money+304A IPC
Burden of proof- patient	Doctor has to proof
Ex- overdose of drug - side effect	Ex-surgeon forget instrument in body

1. Therapeutic Misadventure

An unintended injury or complication resulting from medical treatment, even when the healthcare provider followed standard procedures.

Key Point: No negligence — the harm was accidental despite proper care.

Example: A patient has a rare allergic reaction to a commonly used drug, despite no known allergies or contraindications.

2. Contributory Negligence

When a patient's own negligence contributes to the harm or injury they suffer.

Key Point: Both the patient and possibly the doctor are partly at fault.

Example: A doctor advises a patient to rest after surgery. The patient ignores the advice and resumes heavy lifting, leading to complications.

3. Res Judicata

If a court has already ruled on a malpractice suit and dismissed it, the patient cannot file the same suit again.

4. Res Indicata (less common / often confused with res judicata)

Same as res judicata – once a case is settled, it cannot be reopened.

DECLARATION

Geneva	Hippocratic oath
Ottawa	Child health
Sydney	Definition of death
Tokyo	Torture
Oslo	Therapeutic adventure

Ballistic

FMGE 24

Caliber- distance between two opposite land

How to increase range in a shotgun?

Ans- by choking and rifling

Types of bullets

1. Dum- Dum bullet - terminal end of bullet is chisel out
- maximum damaging capacity



2. Ricochet bullet- strike object and deflected

3. Tandem bullet

➤ Tandem bullet or piggyback bullet:

- Tandem=One behind other.



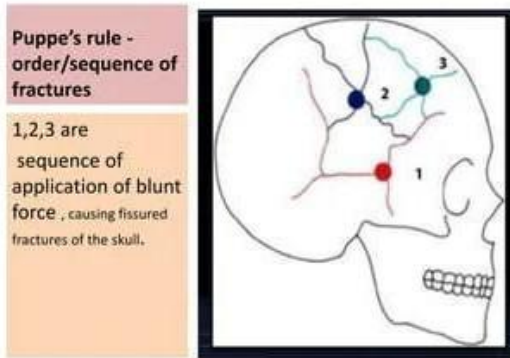
Bullet

Entry wound	Exit wound
Size- small	large
Exception- contact shot- large size Stellate shape	small
Margin- inverted	everted
Tissue color- cherry red color	No color
Abrasion collar	-
Muzzle impression	-
Singing of hair- flame	-
Blackening- burnt gunpowder smoke	-
Tattooing- unburnt gunpowder	-

Stellate shape-



Puppe's rule-when two fractures meet at the same point then the second fracture line never crosses the second fracture line.



Bullet shot

NEETPG 24,FMGE 23,FMGE 2026

	contact	close	near(intermediate)	distant
Muzzle impression	+	—	—	—
Shape of wound	stellate	circular	circular	circular
Burning, tattooing, blackening	Inside track	+	tattooing	—
abrasion, grease, collar	+	+	+	+

Asphyxial death

Lack of O₂

Triad-

C-cyanosis P-petechiae C-congestion

Tardieu spots- petechiae and purpuric rash that develop in the dependant part due to rupture of blood vessels



Types of hanging-

1. Complete- whole body suspended
2. Incomplete- some part of body touching ground
3. Typical- knot tie in occiput area
4. Atypical- knot tie at other place than occiput

Trait	Hanging	Strangulation by ligature
	Suicidal mostly	Homicidal mostly
saliva	Often dribbles out the mouth	Dribbling of saliva is very rare
Ligature mark	Oblique, non continuous	Horizontal and transverse, complete
Thyroid cartilage fracture	Less common	More common
Tardieu spot	+	+
Hyoid fracture	Abduction (outward)	

Postmortem finding of hanging

Ligature mark → most specific finding

Most characteristics AM hanging → dribbling of saliva

Suffocation

purest form of asphyxia

-Cyanosis+, petechiae +, congestion +

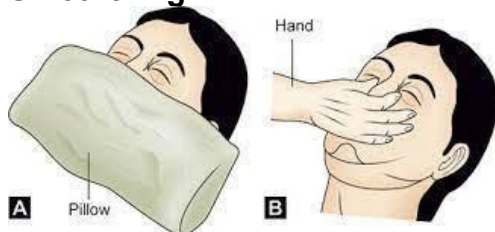
Universal sign of choking



Choking-

Smothering- oral+ nasal both obstruction

NEETPG 24



Traumatic asphyxia- chest fixed due to trauma

Burking- combination of both smothering and traumatic asphyxia

FMGE 23



NEETPG 25

Throttling- multiple penny coins , inward hyoid fracture



Drowning-

FMGE 25

Wet drowning- most common

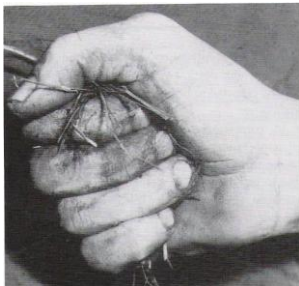
Typical Dry drowning- atypical

Cause of death is laryngeal spasm

Ante mortem findings-

FMGE 24,23

1. Most characteristic- cadaveric spasm



Aspects	Diatom test	Gettler Test
Purpose	Confirms if person was alive during drowning	Determines type of drowning (fresh vs saltwater)
Basis	Diatoms in internal organs = ante-mortem drowning	Chloride level difference in heart blood
Sample	Bone marrow / internal organs	Blood from left & right heart chambers
Freshwater	N/A	Chloride level ↑
Saltwater	N/A	Chloride level ↓

BLAST INJURY

FMGE 25,24

Primary injury	Secondary injury	Tertiary injury
Tympanic membrane > lung > colon	Marshall triad Abrasion Contusion Laceration	Head and spine injury

Characteristic findings of antemortem burn- carbon/soot particle in trachea

High voltage burn-

Lightening-

1. Filigree burn- branching tree pattern
2. Arborescent burn
3. Lichtenberg flower burn



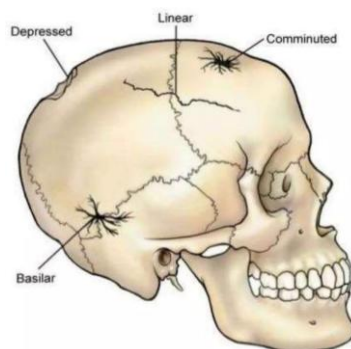
HEAD INJURY

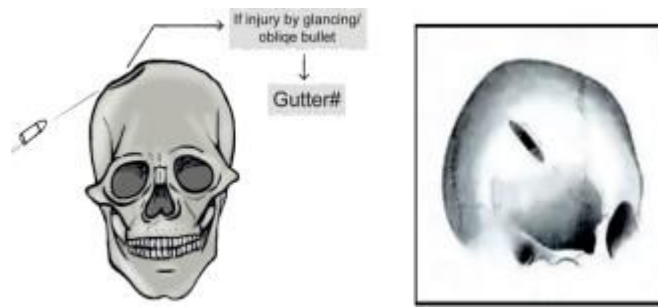
FMGE 25

Most common fracture of skull → fissured/linear fracture

Diastatic fracture → sutural separation

Pond fracture → common in infant





Gutter fracture

Ring fracture - fracture surrounding foramen magnum
Seen from height or falling on fet



Motorcyclist fracture - fracture of the base of skull



Injury caused by sharp weapon

FMGE 25,24,23

1. Incision- maximum dimension is length, knife is parrel to skin



2. Stab

- Knife is perpendicular
- Maximum in depth



3. Laceration- heavy blunt force

Cut laceration- chop wound with axe Cut tissue, crush tissue

Split laceration- bony prominence

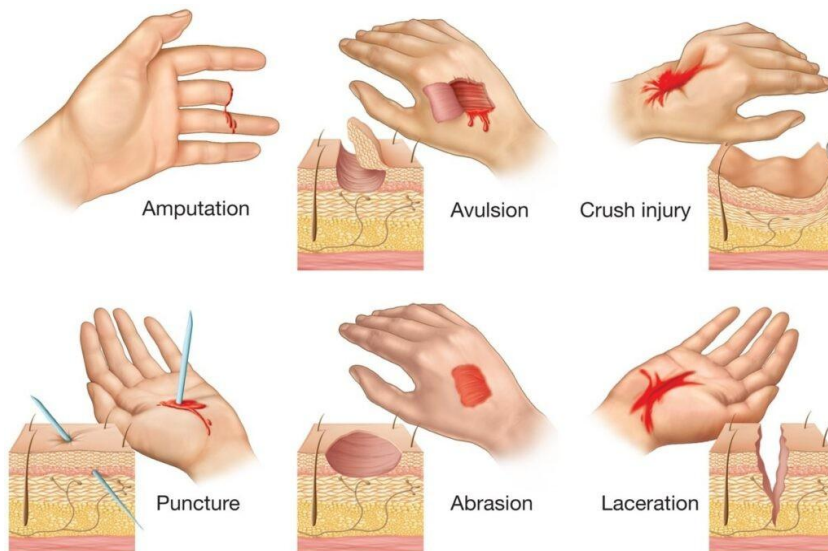
Incised looking, lacerated wound Crush hair follicle seen in lens

4. Avulsion

Detachment of tissue

Flaying - large area is detached, but one point is connected to surface

Degloving - skin or subcutaneous tissue comes out from limb



(laceration - swallow tail , stab - fish tail)

5. self-defense wounds- mc site is extensors of forearm knuckles

MC site of seat belt injury- mesentery

Dashboard- patella #, posterior cruciate ligament

-MC organ damage in blunt injury- spleen

-MC organ damage in penetrating injury- liver

-MC organ damage in underwater blast- intestine

ANTEMORTEM VS POSTMORTEM ABRASION

Postmortem	Antemortem
Site - bony prominences	Site - any site
Intravital reaction - absent	Intravital reaction - present
Scab - less raised	Scab - raised

Artificial bruise	True bruise
Vesicles + Pruritis +	Color change + Vital reaction +

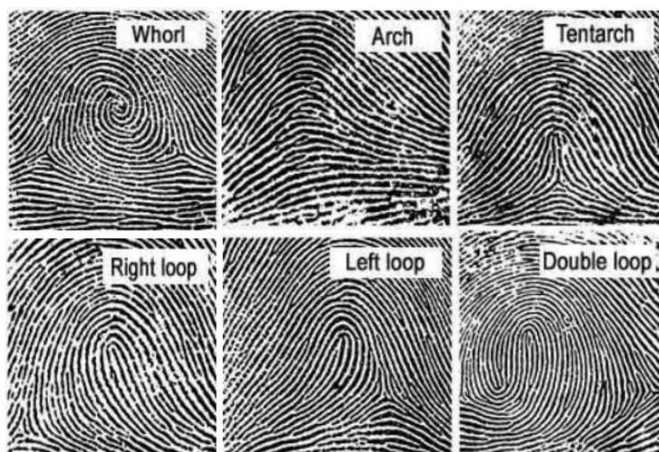
IDENTIFICATION

Best method to identify criminal - fingerprinting

Types of fingerprints

FMGE 2026

Loop → common



Whorl : 2 delta .

Tattoo mark

FMGE 23

Injected in depth of dermis, TILL LYMPH NODE

Superimposition technique - Screening test
Skull and superimposition

FMGE 23

Temporary teeth

FMGE 23

total 20 in number

Lower central incisor is the first tooth to erupts- 6 months

Last to erupt- second molar in 2 yrs

Permanent teeth erupts

FMGE 23

18-25 yrs of age- 32 in number

First molar- 6yrs

Third molar- 17-25yrs

Age estimation using teeth

Gustafson method - useful for > 25 years

Most common - anterior teeth used

Criteria - transparency of root of tooth

THANATOLOGY- STUDY OF DEATH

FMGE 23

POSTMORTEM CHANGES

NEETPG 24,FMGE 24

1. Earliest changes of death- insensibility or loss of voluntary power

2. Length increase after death- relaxation of IVD

3. Eye changes-

k- Kevorkian sign - earliest eye sign

Tache noir- black spots in eye

4. Algor mortis - decrease in temperature

5. Color changes of skin - post mortem lividity/ Post mortem staining/ livor mortis
- absent in pressure area



Rigor mortis- death → stiffening of muscle; primary relaxation is present

In cadaveric spasm primary relaxation is absent.

DECOMPOSITION

seen externally

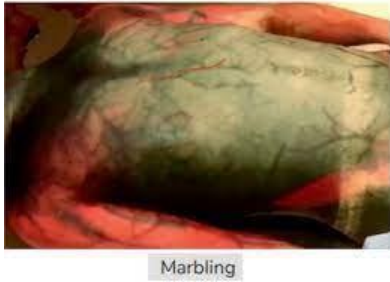
Surest sign of death

Most common bacteria involve is clostridium welchii

First external changes- seen in right iliac fossa → green in color due to sulfhemoglobin

1st internal changes → seen in aortic intima → brown in color

2nd → marbling → chest → superficial vein prominent



Putrefaction

Seen internally

First organ to putrefy- lungs/trachea

Mummification

3 months to 1 yr dry/hot climate

Dehydrated Odorless

Adipocere

Bacteria involve- *Clostridium welchii*

Dry yellow in color

NEETPG 24

Died due to starvation- Most characteristic postmortem finding - gall bladder distention

Infanticide- killing child < 1yr age Neonaticide- kill < 24hrs of neonate Filicide- child killed by parents

Posthumous child- born after death of father

Suppositious child- fictitious child—> never born (blackmailing)

Toxicology

COLOR OF POSTMORTEM STAINING

CO- CHERRY RED COLOR

CN- BRICK RED

H₂S- BLUISH GREEN

HYPOTHERMIA- PINK COLOUR

SPECIFIC ODOR**NEET 2025**

Fruity smell	Ethanol
Rotten eggs smell	H ₂ S
Fishy odor	Aluminium phosphide
Bitter almonds	Cyanide
Burnt rope	Cannabis
Garlic smell	Arsenic , phosphorus

Strychnine**NEETPG 23**

-Inhibit glycine



-act on anterior horn of spinal cord

-look like tetanus, lock jaw

-mc position- opisthotonus

Cause of death- asphyxia

Antidote- diazepam

Gastric lavage is indicated in poisoning of-

Morphine Cocaine Amphetamine Barbiturate

Gastric lavage is contraindicated in corrosive poisoning- except carbolic acid

Coma cocktail antidote

D-dextrose N-naloxone B1-thiamine

Datura**NEETPG 24,23,FMGE 24**

Most common found in road side and railway

Anticholinergic

9- D's

Dry and hot skin

Delirium

Dermatitis

Dilated pupil Dysphagia Delusion Diplopia

Dilated vessels Drunken gait

Antidote- physostigmine

-pilocarpine



Datura alba



datura nigro



datura stromonium

Cannabis sativa

Cns stimulant

Smelling like burning rope

Antidote- diazepam



Opium

Pin point pupil

Depress all centers except vomiting and sweating

MORPHINE

M-miosis

O-orthostatic hypotension

R-respiratory depression

P-pain relief

H-histamine increase → bronchospasm

I-increase intracranial tension

N-naloxone

E-euphoria

Antidote of choice- naloxone

Oral Antidote- naltrexone



Oleander poisoning

Yellow oleander



Acute principle - cerebrin, nerifolin

Pink oleander - nerium odorum

Active principle - oleandrin

FMGE 23



MOA

Action is like digitalis

Inhibition of NaK ATPase channel leads to bradycardia/tachycardia, AV block, $\uparrow K^+$

Treatment

Antidote $\rightarrow$ digibind

For AV block - atropine

Snakes

Viper - Vasculotoxic

Three rows of brown spots



NEETPG 23, FMGE 24

Sea snake- myotoxic



Common krait- respiratory muscle paralysis



King cobra



Treatment

Follow RIGHT strategy

R- reassure

I-immobilize

GH- go to hospital

T- tell appropriate history

ASV+neostigmine+atropine → cobra snake

ASV+FFP → viper bite

Phosphorus poisoning

Garlicky smell

Asterixis tremor

Phossy jaw

Antidote- CuSO_4 , vit k

Mercury poisoning

c/f- acrodynia

Brown malt reflex Erethism

Antidote- BAL

Lead poisoning

Cause sideroblastic anemia- microcytic hypochromic anemia

Chronic- Plumbism

A-anemia, amenorrhea

B-blue dot-rbc - basophilic stippling

Blue line in gums- Bartonian line

C- constipation

D- dry belly ache, wrist, and foot drop

Earliest most consistent symptoms- face pale



Antidote- EDTA

Lead inhibit-

1.Ferrochelataase

2.ALA dehydratase

3.Coproporphyrinogen

Arsenic poisoning

c/f-Aldrich mees line , Garlicy odor in acute poisoning

Red velvety mucosa-postmortem finding

Rain drop pigmentation- postmortem finding



NEETPG 25,24



Excessive pigmentation of palm and sole Neuropathy

Antidote-BAL

Note- preservative use to preserve viscera- aconite

ZASKO'S PHENOMENON : Tendon reflex occurring after death. (NEET 2025).

QUOUD HOC : impotency in male , unable to engage in sexual intercourse with one specific individual.

Genetic marker in DNA FINGERPRINTING : Short tandem repeats is used for identification and comparison among family members

MAXIMUM AGE OF FUSION

BONE /JOINT	MAX AGE OF FUSION
Medial end of clavicle	20-22 years
Wrist joint	18-19 years
Pelvis	20-21 years
Shoulder joint	17-18 years
skull	45 to 50 (lambdoid suture)

.

*Putrefaction of organs :

Order : Brain, heart ,uterus , then skin

Postmortum hypostasis

Poisoning	Staining
CO	Cherry red
CN	BRICK RED
HYPOTHERMIA	BRIGHT PINK
OPIUM	GREYISH /BLACK
PHOSPHORUS	DARK BROWN
HYDROGEN SULPHIDE	BLUISH GREEN